# MILITARY SPECIFICATION

## GRENADE, HAND, RED SMOKE, RS, M48

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers one type of training grenade.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| QQ-S-781 | - | Strapping, Steel, Flat and Seals. |
| TT-W-572 | - | Wood Preservative, Water Repellent. |
| PPP-B-26 | - | Bag, Plastic. |
| PPP-B-601 | - | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-F-320 | - | Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes. |
| PPP-T-97 | - | Tape, Pressure Sensitive Adhesive, Filament Reinforced. |

MILITARY

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of. |
| MIL-B-117 | - | Bags, Sleeves and Tubing, Interior Packaging. |
| MIL-B-2427 | - | Boxes, Ammunition Packing; Wood, Nailed. |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |



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MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

D13-25-71 - Grenade, Hand, Red, Smoke, RS, M48

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D13-25-71 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the grenade shall conform to the specifications and drawings listed on Drawing D13-25-71 and on subsidiary drawings.

3.1.3 Red smoke mixture. The volatile content of the final mix when loaded into munition shall be a miximum of 0.5 percent by weight when tested in accordance with 4.4.4.1 (see 6.4).

3.2 Assembly. The grenade shall be assembled as shown on Drawing D13-25-71.

3.3 Functioning. The grenade shall function as follows when tested in accordance with 4.4.4.2:

(a) The fuze shall function and cause the fill to ignite and emit a red smoke cloud under continuous pressure for no less than 5 seconds and for no more than 25 seconds. Flaming (see 6.5) shall not occur for more than 10 percent of the emission time.

(b) The time measured from the audible functioning of the fuze to first smoke emission under pressure shall be no less than 0.5 seconds and no more than 5.0 seconds. The primer shall function prior to impact of grenade on the ground.



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(c) The grenade shall move/skitter on the ground during at least 50 percent of the smoke emission time when functioned on a hard level surface.

3.4 Immersion. The grenade shall not have any evidence of leakage after water immersion and shall conform to the requirements of 3.3 when tested in accordance with 4.4.4.2.

3.5 Preproduction. Prior to the start of regular production, a preproduction lot of grenades shall be produced in accordance with this specification for examination and test.

3.6 Workmanship. The grenade shall be free from contamination (oil, grease or foreign matter) and damage such as cracks, breaks or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier in responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 250 grenades shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.



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4.3.2 Inspection procedure.

4.3.2.1 For examination and test. The preproduction lot shall be sampled, inspected and tested in accordance with table I.

Table I. Preproduction sampling, inspection and acceptance/rejection criteria

| Quantity | Requirement | Test | Accept | Reject |
|---|---|---|---|---|
| 1/2 pound red smoke mix from each batch (see 6.3) | 3.1.3 | 4.4.4.1 | 0 | 1 |
| 65 grenades | 3.2 | 4.4.3.4 | 1 | 2 |
| | 3.3 | 4.4.4.2 | *1 | 2 |

*Any grenade exhibiting one or more of the defects listed in 4.4.4.2 shall be considered a reject. No one grenade will be counted as a reject more than once, regardless of how many defects that one grenade exhibits.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. However, no more than one interfix number of each component (fuzes, body assemblies and red smoke mixture) from one manufacturer shall be contained in any one lot of assembled grenades.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test.

(a) Grenade. Sampling shall be conducted in accordance with MIL-STD-105.

(b) Red smoke mixture. A sample of 1/2 pound of red smoke mixture shall be taken from each batch (see 6.3) prior to filling the grenade.

4.4.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-4.



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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive test.

(a) Grenade. The sample grenades and the level A preparation for delivery shall be examined in accordance with the classification of defects and with MIL-STD-105.

(b) Red smoke mixture. The sample of mixture shall be tested in accordance with 4.4.4.1.

4.4.3.2 For destructive test. The sample grenades shall be tested in accordance with 4.4.4.2 and with MIL-STD-105 using an AQL of 2.5 percent defective. Any grenade exhibiting one or more of the defects listed in 4.4.4.2 will be considered a reject. No one grenade will be counted as a reject more than once, regardless of how many defects that grenade exhibits.

4.4.3.3 Critical defects. Each grenade in the lot shall be examined for its critical characteristic listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Grenade, hand, red smoke, RS, M48 (Drawing D13-25-71).

| Categories | Defects |
|---|---|
| Critical: | |
| 1 | Safety pin missing |
| Major: | AQL 1.0 percent defective |
| 101 | Component missing |
| 102 | Component incorrectly assembled |
| 103 | Marking missing or incorrect |
| 104 | Filling weight incorrect |
| 105 | Emission port tape missing, mis-positioned, wrinkled or loose |
| 106 | Workmanship (3.6) |



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(b) Preparation for delivery (section 5).

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | AQL 4.0 percent defective |
| 101 | Packaging loose |
| 102 | Packing loose |
| 103 | Container incorrect |
| 104 | Barrier bag incorrectly sealed |
| 105 | Barrier bag ruptured |
| 106 | Marking missing or illegible |
| 107 | Marking incorrect |

4.4.4 Tests. The tests shall be conducted as follows:

4.4.4.1 Volatile content. Volatile content shall be determined by placing into a tared weighing dish five (5) to ten (10) grams of mix weighed to the nearest 0.1 milligram (0.0001 gram). Place test sample in a vacuum desiccator with desiccant and apply a vacuum of 5 to 10 millimeters of mercury for 2 hours with a vacuum pump. Remove vacuum slowly, so sample will not be blown from pan, and reweigh. Weight loss is calculated as percent volatiles.

$$\text{Percent volatiles} = \frac{\text{weight loss}}{\text{sample weight}} \times 100$$

4.4.4.2 Functioning. The grenade shall be immersed in water having a temperature of 0° to 10° F above the temperature of the grenade, to a depth of two (2) inches for 30 ± 5 minutes. At the end of the immersion period the grenade shall be wiped dry and functioned within 1 hour. To function, the grenade shall be armed and thrown at least 50 feet onto a hard level surface. The grenade shall be observed for defects as follows:

(a) Dud (fuze fails to function, or red smoke pyrotechnic mix fails to ignite)
(b) Failure to emit a continuous cloud for the required time (5 to 25 seconds)
(c) Delay incorrect (0.5 to 5.0 seconds)
(d) Flaming (see 6.5)
(e) Skittering (see 3.3(c))
(f) Hangfire (primer functions after impact of grenade on ground)



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## 5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. Four grenades shall be packaged in a two-section foamed plastic container conforming to Drawing D13-25-81. The two sections shall be secured together with two bands of 3/4-inch-wide pressure-sensitive tape conforming to PPP-T-97, type I, applied girthwise in the provided grooves. The container shall then be placed within a barrier bag measuring 15 inches wide by 15 inches long, conforming to type I, Class E of MIL-B-117, in accordance with method IA-8 of MIL-P-116, except that the backing material shall be of spunbonded polyolefin or nylon. The barrier bag shall be closed by heat sealing and evacuated air volume within the sealed bag. The barrier bag shall be subsequently tested for leakage in accordance with the leak test requirements of MIL-P-116.

5.2 Packing, level A. Five packages of greande (20 grenades) packaged in accordance with 5.1 shall be packed in an overseas type or Class box conforming to PPP-B-601, PPP-B-621 or type I, Class 2, Grade A of MIL-B-46506. Inside dimensions in inches of the box shall be 22-3/4 long by 8-3/4 wide by 9 deep. Prior to loading, the box or all wood components thereof, shall be immersed for a minimum of one minute in wood preservative conforming to Composition A of TT-W-572. Prior to packing the item in the boxes treated with wood preservative, the box shall meet the dryness requirement for preservative as specified in MIL-B- 2427. Fiberboard pads conforming to type CF, variety SW, Grade W5c of PPP-F-320, shall be used between the sides, ends, top and bottom of the box and contents. Additional pads shall be used as required to make a tight pack. The box shall be closed in accordance with the appendix of the applicable box specification.

5.3 Palletization. When required by the contract or order, grenades packed as specified in 5.2, shall be palletized in accordance with Drawing D13-25-75.

5.4 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129 to include "LOT NUMBER". Shipping containers shall be marked in accordance with Drawing D13-25-74.

## 6. NOTES

6.1 Intended use. The M48 grenade is intended for use as a training item for personnel in preparation for the use of the M47 grenade.



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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) The shipment group size and marking (when applicable).
(c) Pertinent identification marking.
(d) Applicable additional marking.
(e) Shipment priority number, if applicable.
(f) Transportation control number, project code and required delivery date.
(g) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been subjected to some chemical or physical mixing process intended to yield a substantially uniform product.

6.4 Caution. Precautions should be taken to prevent fires during preparation of the mixtures and while assembling grenades. Due to the hazard from exposure to organic dyestuffs, a dust respirator, coveralls and gloves should be worn when handling these materials. Caution should also be exercised during the functioning test as a hazard exists from flying hot particles and possible detonation resulting from incorrect mixture.

6.5 Flaming. Flaming is defined as a malfunction in which there is visible flame during burning of the mixture.

6.6 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description EA-G-1015C dated 28 August 1972.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A-106



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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